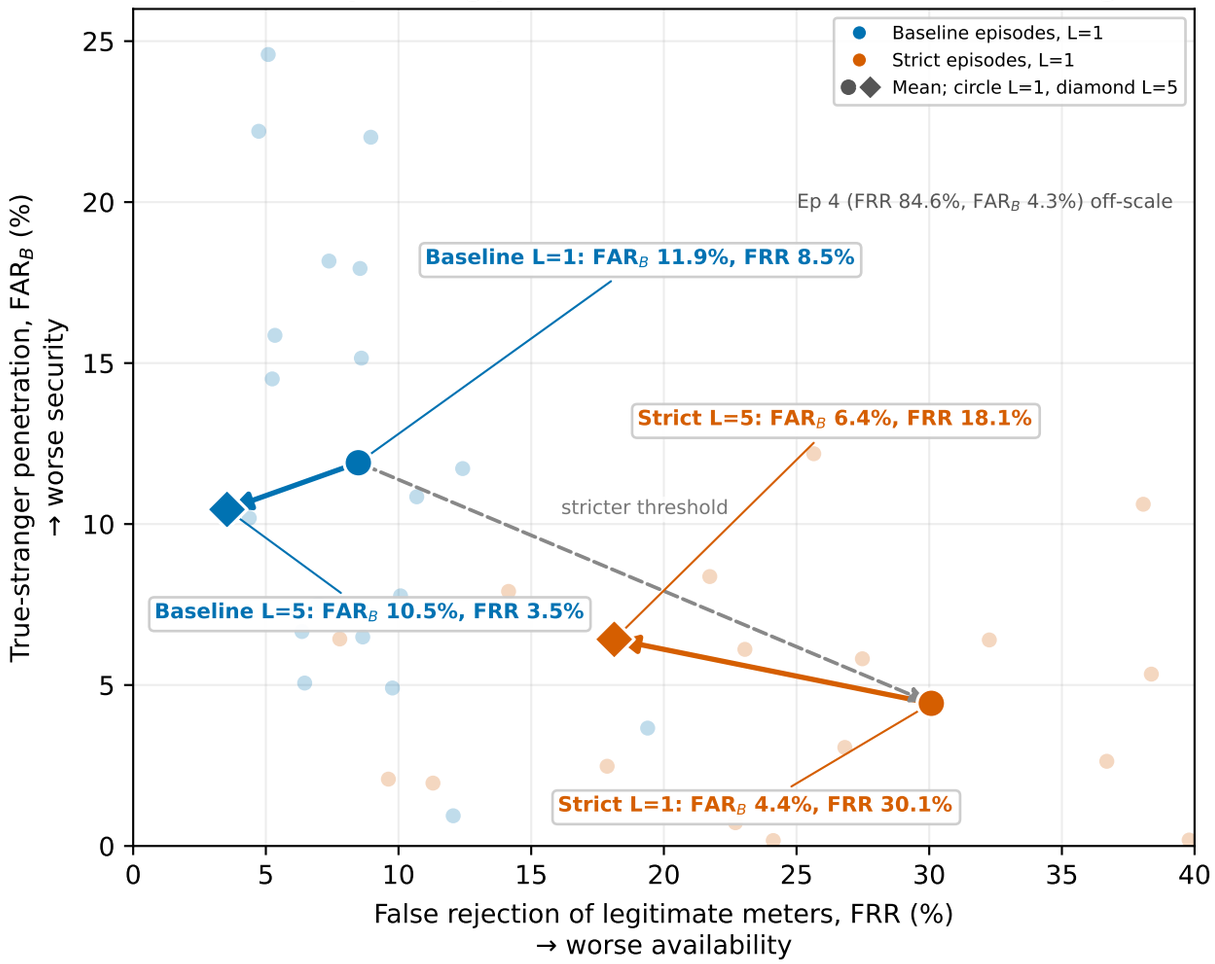
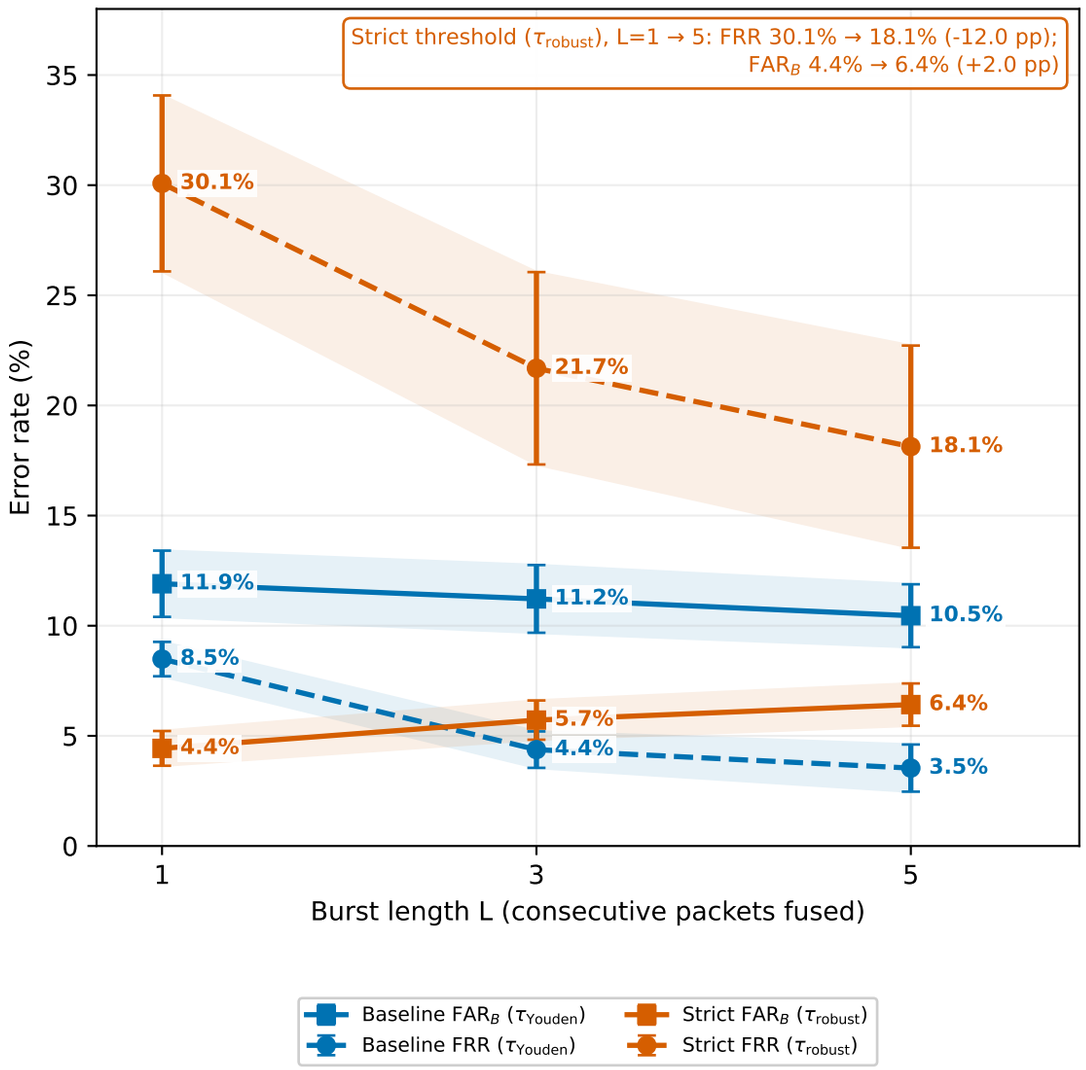

(a) Mean operating points across threshold and burst length
Baseline episodes, L=1
Strict episodes, L=1
Mean; circle L=1, diamond L=5
Ep 4 (FRR 84.6%, FAR$_B$ 4.3%) off-scale
Baseline L=1: FAR$_B$ 11.9%, FRR 8.5%
Strict L=5: FAR$_B$ 6.4%, FRR 18.1%
stricter threshold
Baseline L=5: FAR$_B$ 10.5%, FRR 3.5%
Strict L=1: FAR$_B$ 4.4%, FRR 30.1%
True-stranger penetration, FAR$_B$ (%)
→ worse security
False rejection of legitimate meters, FRR (%)
→ worse availability
0
5
10
15
20
25
30
35
40


(b) Burst fusion lowers FRR; FAR$_B$ rises modestly at strict threshold
Strict threshold (τ_{robust}), L=1 → 5: FRR 30.1% → 18.1% (-12.0 pp);
FAR$_B$ 4.4% → 6.4% (+2.0 pp)
30.1%
21.7%
18.1%
11.9%
11.2%
10.5%
8.5%
4.4%
5.7%
6.4%
4.4%
3.5%
Error rate (%)
Burst length L (consecutive packets fused)
Baseline FAR$_B$ (τ_{Youden})
Baseline FRR (τ_{Youden})
Strict FAR$_B$ (τ_{robust})
Strict FRR (τ_{robust})
0
5
10
15
20
25
30
35
1
3
5